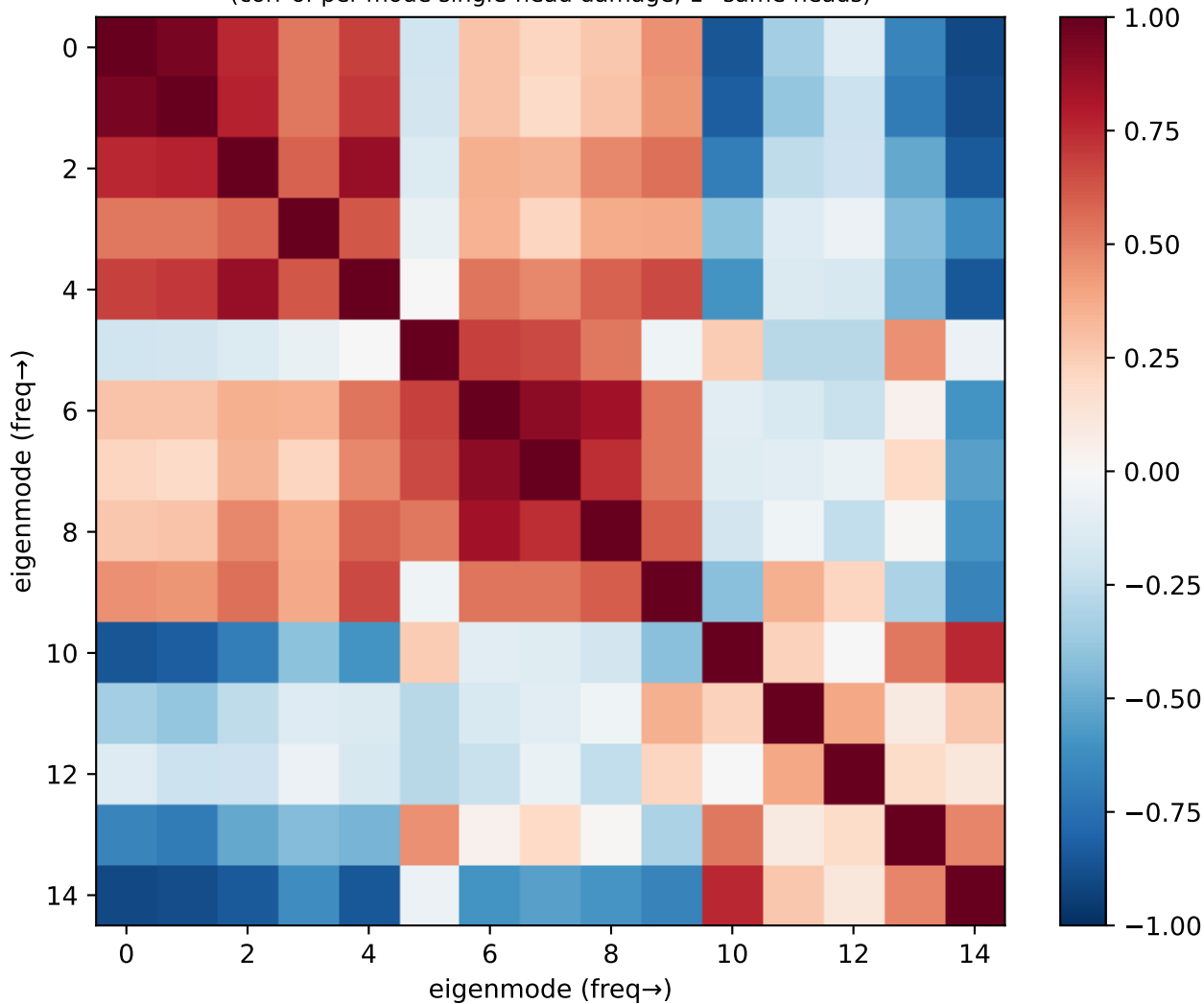
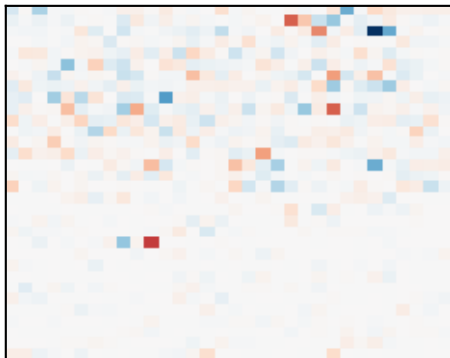


Llama grid: eigenmode circuit-overlap
(corr of per-mode single-head damage; 1=same heads)

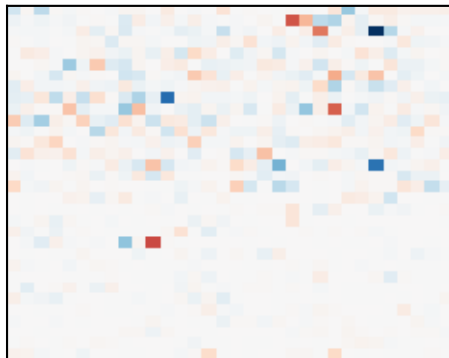


Llama grid: single-head damage per eigenmode (red=head builds it)

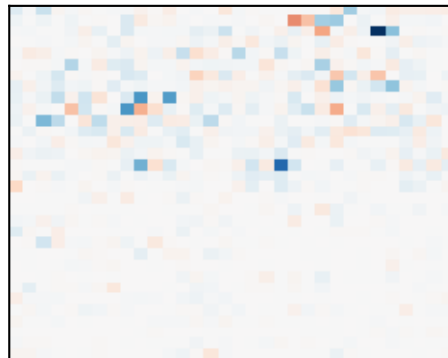
m1 $\lambda 0.6$ p0.16



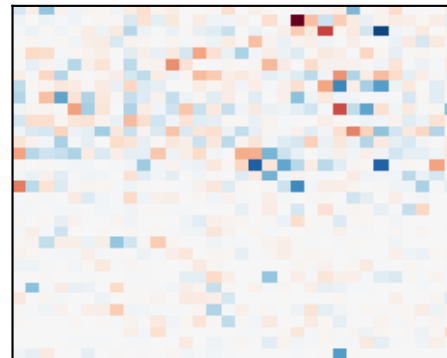
m2 $\lambda 0.6$ p0.13



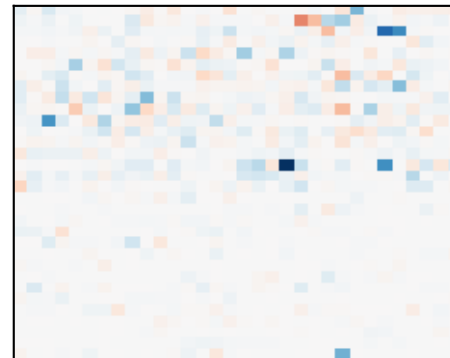
m3 $\lambda 1.2$ p0.05



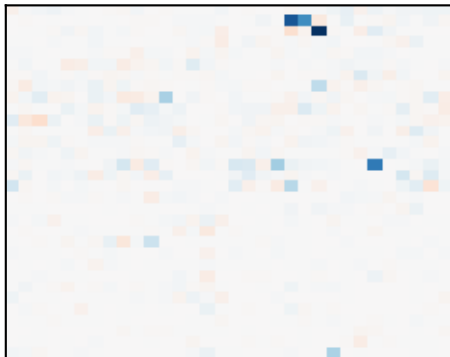
m4 $\lambda 2.0$ p0.04



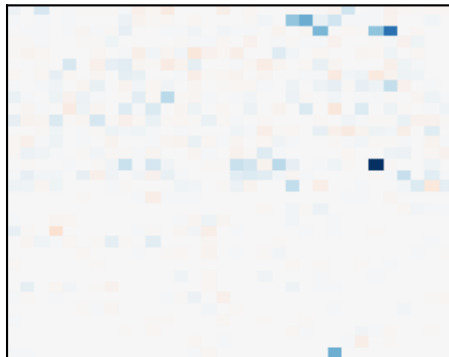
m5 $\lambda 2.0$ p0.02



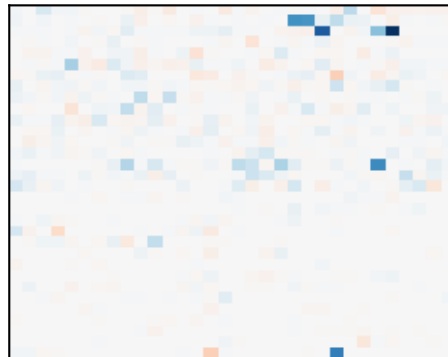
m6 $\lambda 2.6$ p0.03



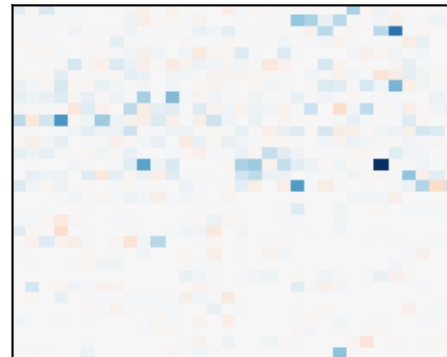
m7 $\lambda 2.6$ p0.02



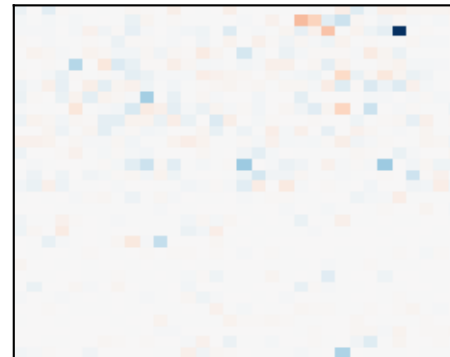
m8 $\lambda 3.4$ p0.02



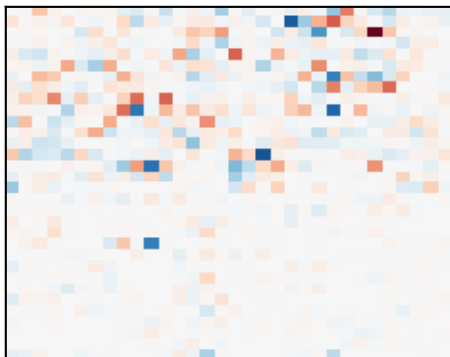
m9 $\lambda 3.4$ p0.02



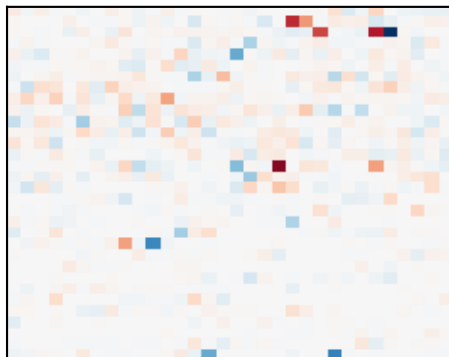
m10 $\lambda 4.0$ p0.03



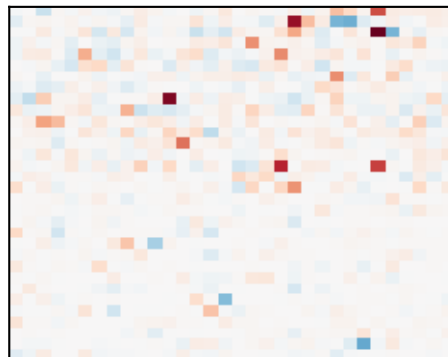
m11 $\lambda 4.0$ p0.04



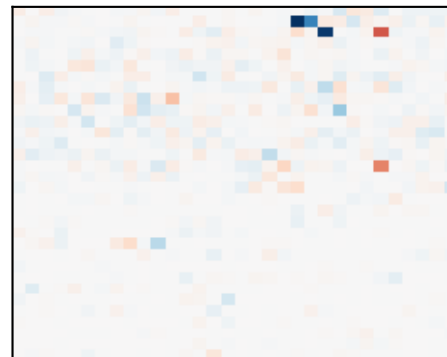
m12 $\lambda 4.0$ p0.04



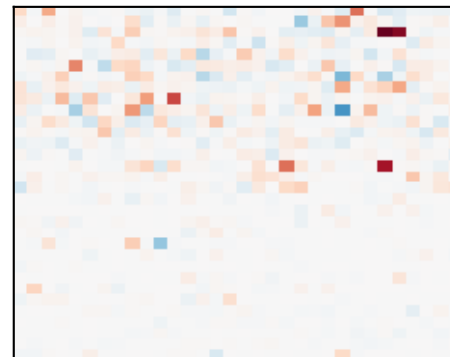
m13 $\lambda 5.4$ p0.05



m14 $\lambda 5.4$ p0.06



m15 $\lambda 6.8$ p0.28



Llama – grid: which heads build which structure

CIRCUIT 1 – low-freq (no simple name) / x-position + y-position
modes [1, 2, 3] (λ 0.6, 0.6, 1.2), total power 0.34
top heads: L21H10, L9H23, L1H20

CIRCUIT 2 – parity/checkerboard
modes [15] (λ 6.8), total power 0.28
top heads: L2H26, L2H27, L14H26

CIRCUIT 3 – high-freq (no simple name)
modes [14] (λ 5.4), total power 0.06
top heads: L2H26, L14H26, L8H11

CIRCUIT 4 – high-freq (no simple name)
modes [13] (λ 5.4), total power 0.05
top heads: L2H26, L8H11, L1H20

Circuit 1 vs Circuit 2 cross-correlation: -0.88 (anti-correlated → compete for variance)